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The Courageous Formulation

Austere Quantum Mechanics

The attitude inculcated into young students by modern quantum mechanics textbooks has been compactly summarized by physicist N. David Mermin as "Shut up and calculate!" Mermin himself wasn't advocating such a position, but others have. Every decent physicist spends a good deal of time calculating things, whatever their attitude toward quantum foundations might be. So really the admonition could be shortened to simply "Shut up!"*

It wasn't always thus. Quantum mechanics took decades to piece together, but rounded into its modern form around 1927. In that year, at the Fifth International Solvay Conference in Belgium, the world's leading physicists came together to discuss the status and meaning of quantum theory. By that time the experimental evidence was clear, and physicists were at long last in possession of a quantitative formulation of the rules of quantum mechanics. It was time to roll up some sleeves and figure out what this crazy new worldview actually amounted to.

The discussions at this conference help set the stage, but our goal here isn't to get the history right. We want to understand the physics. So we'll sketch out a logical path by which we will be led to a full-blown scientific theory of quantum mechanics. No vague mysticism, no seemingly ad hoc rules. Just a simple set of assumptions leading to some remarkable conclusions. With this picture in mind, many things that might otherwise have seemed ominously mysterious will suddenly start to make perfect sense.